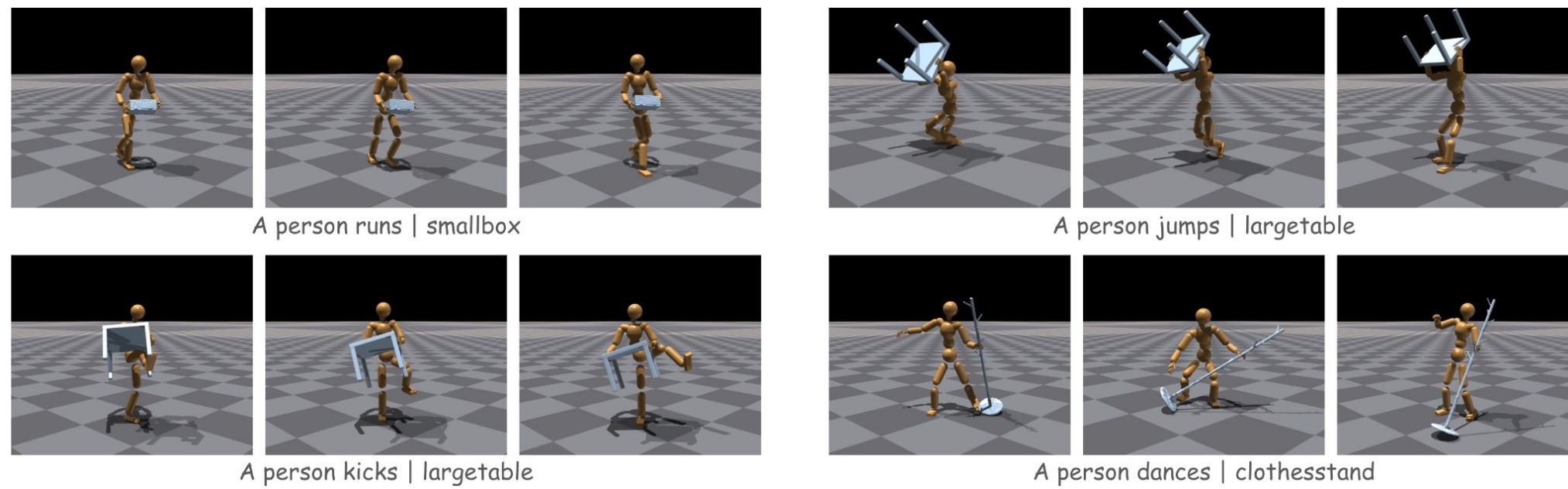




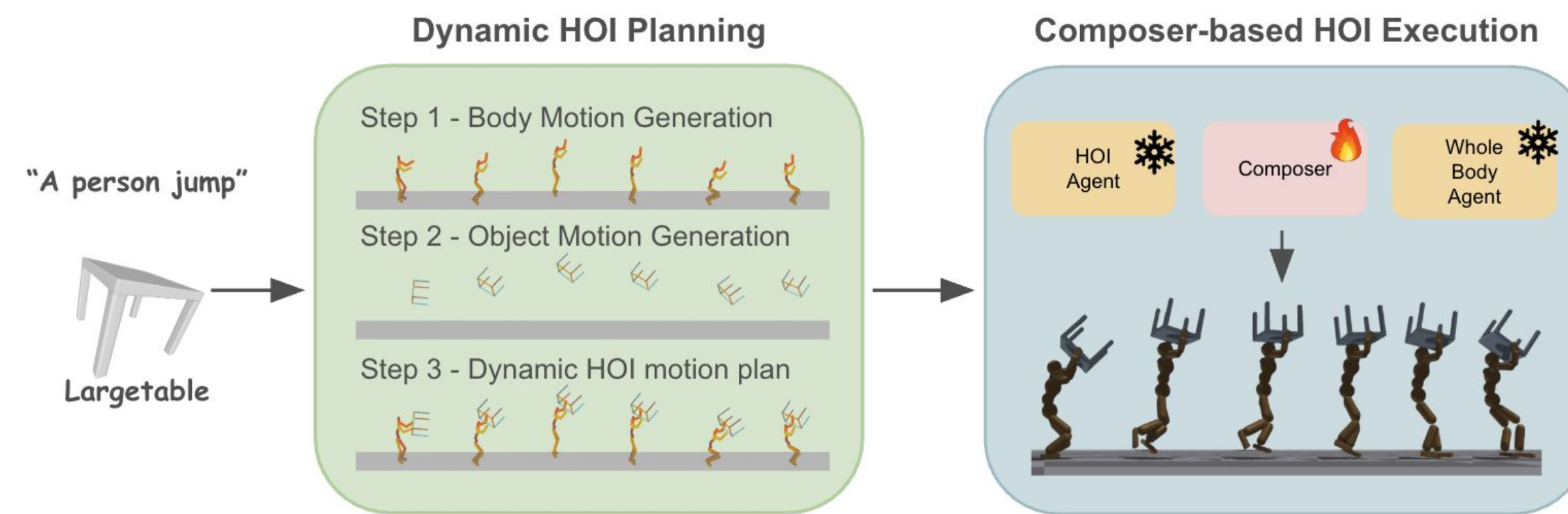
Introduction



run, jump, kick, dance while interacting with an object

- Plan and execute physically plausible, **dynamic** and **contact-rich Human-Object Interaction (HOI)**
- Existing datasets and agents handle either HOI or dynamic motions, but **not both simultaneously**

Overview



- Dynamic HOI Planning:** Plan dynamic motions with consistent hand-object contact
- Composer-based HOI Execution:** Efficiently learn dynamic HOI control by blending pre-trained experts

Quantitative Results

Method	HOI	Physics	HOI QUALITY		PHYSICAL PLAUSIBILITY				MOTION QUALITY	
			$C_{\%} \uparrow$	$C_{\text{cons}} \uparrow$	Pene _{obj} $\downarrow$	Skate $\downarrow$	Float $\downarrow$	Jitter _{pos} $\downarrow$	R-Prec $\uparrow$	Diversity $\uparrow$
MDM			—	—	8.18	0.133	29.3	9.43×10^4	0.374	7.61
HOI-Diff	✓		0.285	18.2	—	0.633	17.8	1.56×10^4	0.257	4.66
DAViD	✓		0.848	10.9	4.842	0.261	20.4	6.69×10^4	0.310	6.70
Ours _p	✓		1.000	0.906	4.196	0.217	30.1	5.93×10^4	0.332	5.56
Ours _{p+E}	✓	✓	0.999	2.95	0.009	0.786	9.95	4.53×10^4	0.316	3.97

Generation Quality

(Ours_p: planning only / Ours_{p+E}: full planning-execution pipeline)

- Ours_p achieves the **best HOI Quality** (best $C_{\%}$ and C_{cons})
- Ours_{p+E} achieves the **best physical plausibility** (best Pene_{obj} and Float, 2nd Jitter_{pose})

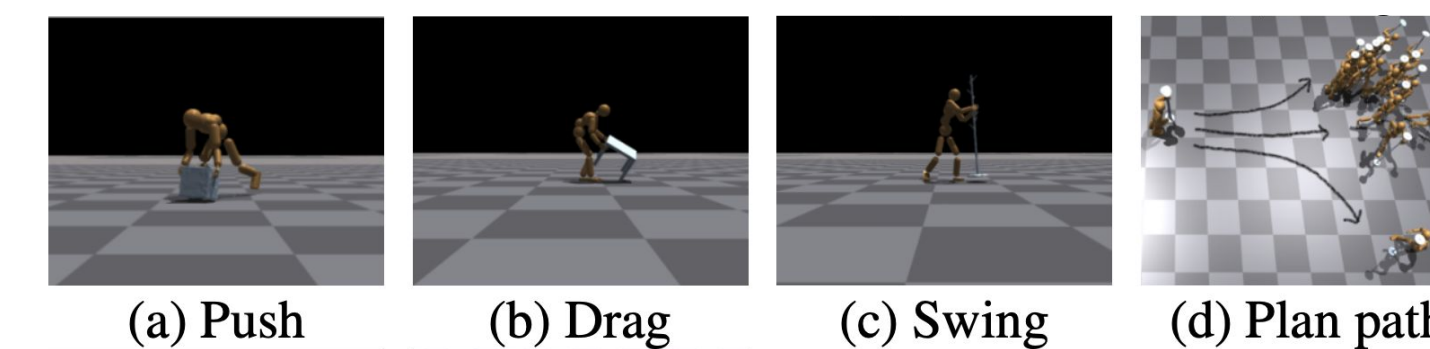
Method	SR $\uparrow$	T	D $\uparrow$	$E_{\text{HOI}} \downarrow$	Jitter _{DoF} $\downarrow$
PPO	0.227	75	0.859	59.098	0.424×10^3
PHC [26]	0.397	—	1.166	12.965	1.350×10^3
PHC _R	0.313	15	1.834	26.024	1.762×10^3
InterMimic [56]	0.376	—	2.407	21.379	0.436×10^3
InterMimic _R	0.310	15	2.324	34.675	0.336×10^3
InterMimic _{FT}	<u>0.526</u>	75	7.37	8.635	0.370×10^3
Ours	0.591	23	<u>4.607</u>	<u>11.667</u>	0.359×10^3

Execution Performance

(Ours_{MLP}: except PCA subspace exploration)

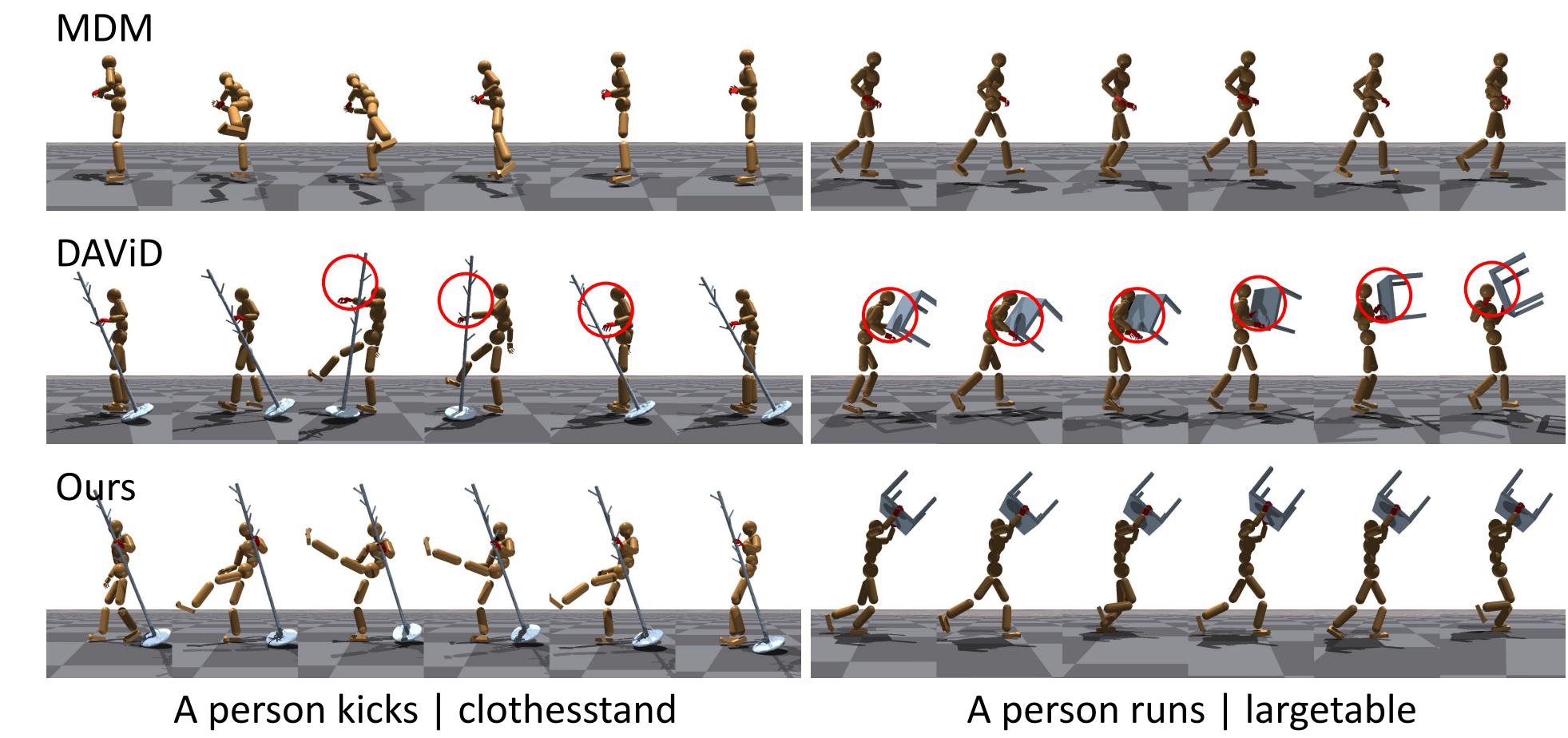
- Ours achieves the **highest success rate**
- Converges **3× faster** than fine-tuning InterMimic
- Ours blending method achieved best SR, D, and E_{HOI}

Scalability



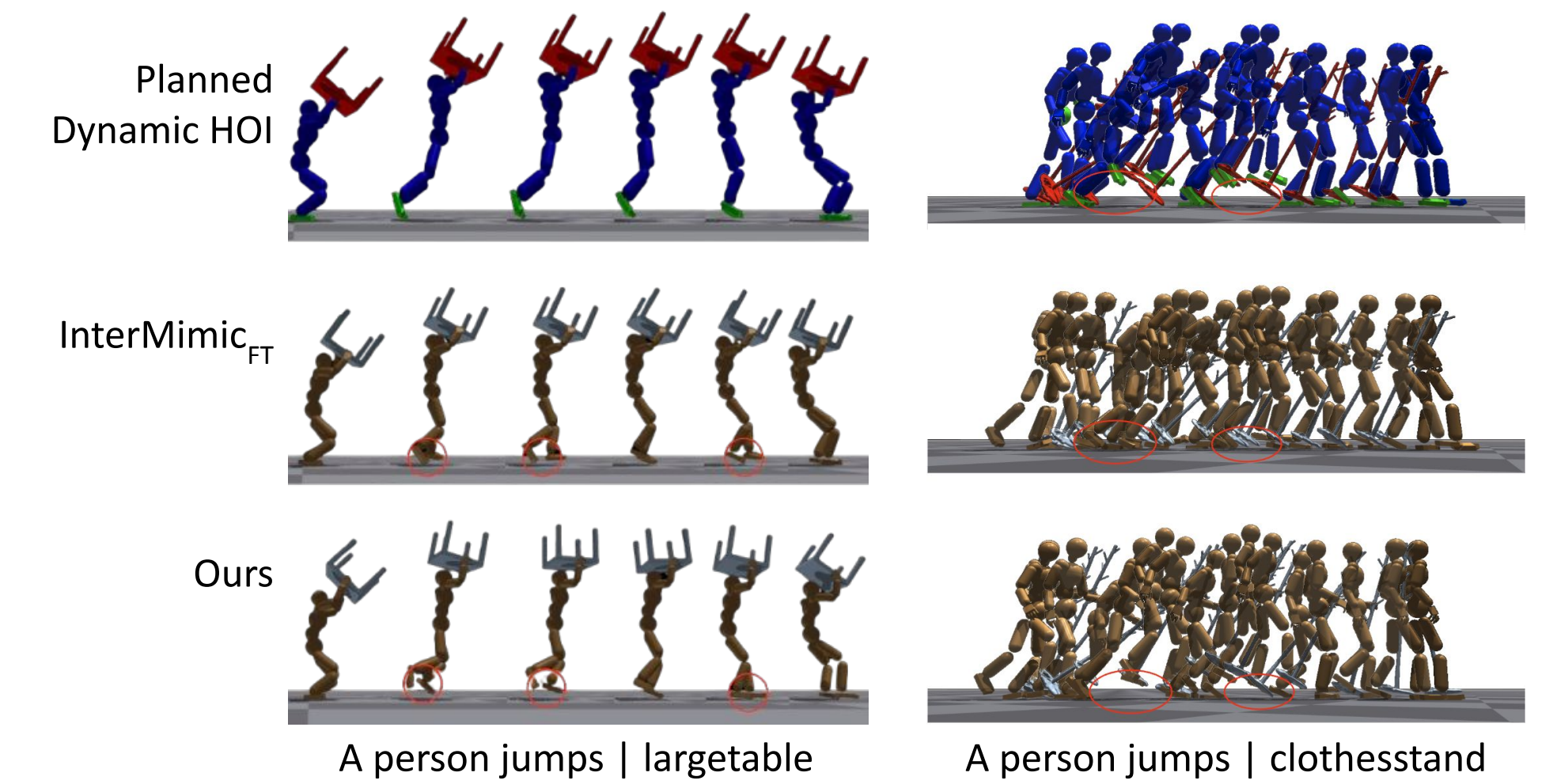
- Diverse object geometries (FullBodyManip)
- Diverse motion styles (push, drag, swing)
- Plug-and-play application — e.g., MDM → GMD enables path planning

Qualitative Results



Comparison of HOI Planning

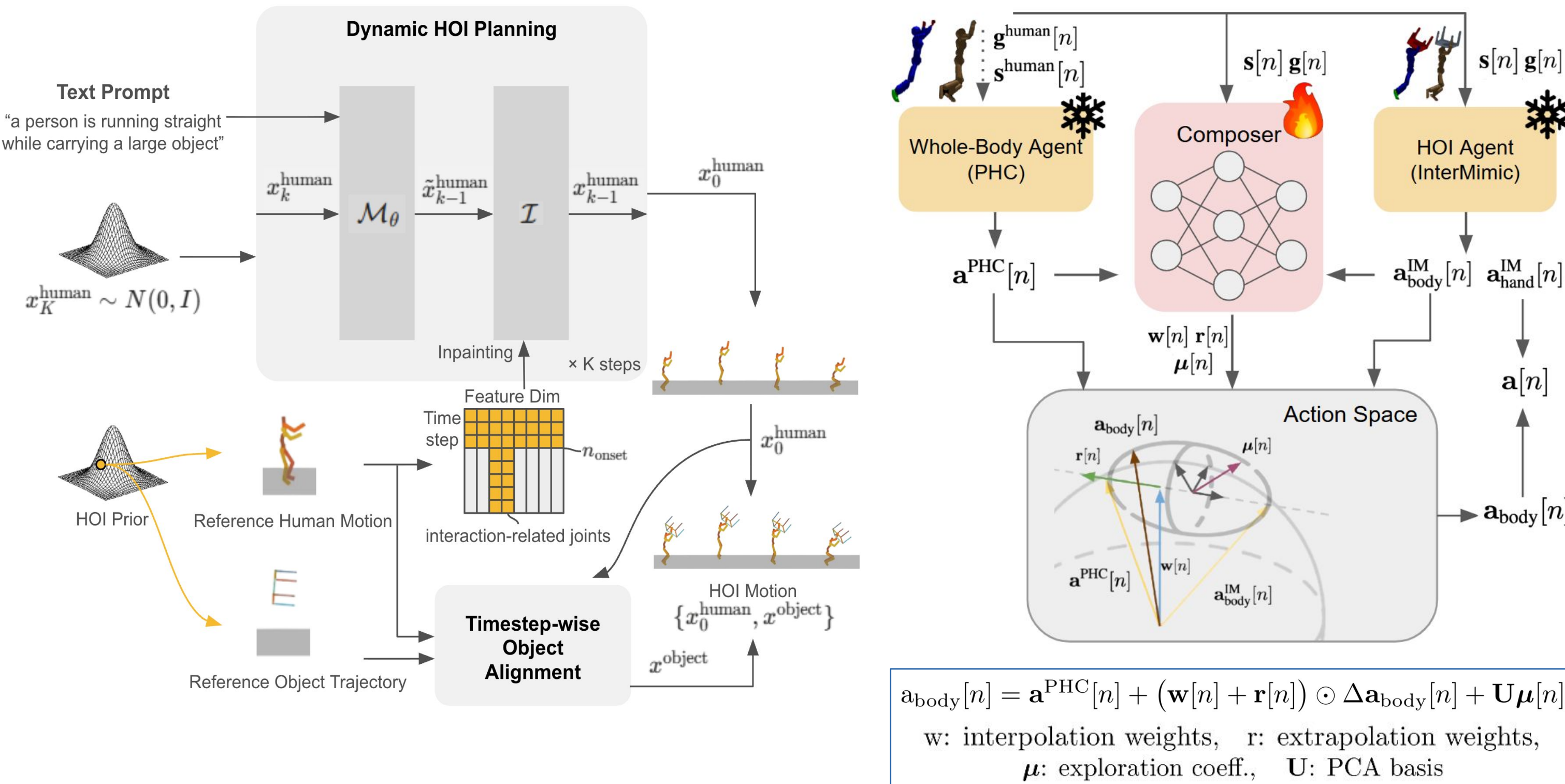
- Ours accurately follows the text-aligned **dynamic motion** while maintaining **consistent hand-object contact**



Comparison of HOI Execution

- Ours successfully executes the planned dynamic HOI while maintaining **motion style** and **object interaction**
- InterMimic_{FT} collapses to a local minimum — shuffling instead of two-foot takeoff

Methodology



Dynamic HOI Planning

- Inpaint HOI prior (FullBodyManip) into MDM during diffusion sampling
- Fix interaction-related joints after contact onset; text-aligned motion inpainting for the others
- Compute object trajectories from hand poses

Composer-based HOI Execution

- Composer predicts per-DoF inter-/extrapolation weights
- PCA-based low-dim subspaces for further exploration

$$\mathbf{a}_{\text{body}}[n] = \mathbf{a}^{\text{PHC}}[n] + (\mathbf{w}[n] + \mathbf{r}[n]) \odot \Delta \mathbf{a}_{\text{body}}[n] + \mathbf{U} \boldsymbol{\mu}[n]$$

$\mathbf{w}$: interpolation weights, $\mathbf{r}$: extrapolation weights, $\boldsymbol{\mu}$: exploration coeff., $\mathbf{U}$: PCA basis

Contact

Sanghyeok Nam
KAIST, Republic of Korea
Email: sang990701@kaist.ac.kr
Website: <https://www.linkedin.com/in/sanghyeok-nam-a05461391>

References

- N. Mahmood et al., "AMASS: Archive of Motion Capture as Surface Shapes," ICCV, 2019.
- J. Li et al., "Object Motion Guided Human Motion Synthesis," ACM TOG, 2023.
- G. Tevet et al., "Human Motion Diffusion Model," ICML, 2023.
- X. Peng et al., "HOI-Diff: Text-Driven Synthesis of 3D Human-Object Interactions Using Diffusion Models," CVPR Workshop on HuMoGen, 2025.
- H. Kim et al., "DAViD: Modeling Dynamic Affordance of 3D Objects Using Pretrained Video Diffusion Models," ICCV, 2025.
- Z. Luo et al., "Perpetual Humanoid Control for Real-time Simulated Avatars," ICCV, 2023.
- S. Xu et al., "InterMimic: Towards Universal Whole-Body Control for Physics-Based Human-Object Interactions," CVPR, 2025.
- K. Karunaratnan et al., "Guided Motion Diffusion for Controllable Human Motion Synthesis," ICCV, 2023.